

Department of Mathematical Sciences

Rajiv Gandhi Institute Of Petroleum Technology, Jais

End Semester Examination (AY 2023 - 24 / Even / Sem - II)

B.Tech. in M&C

Engineering Practices in Mathematics and Computing (MA 122)

Duration : 3 hours

May 2, 2024

Total Marks : 50

Instructions.

- i) Attempt all the questions using MATLAB ii) Calculator is NOT allowed. iii) Symbols have their usual meanings.

1. State whether the following statements are True or False: (1 × 5)

- i) Indentation is required in MATLAB to define code blocks. τ
ii) The eye() function in MATLAB creates a diagonal matrix with ones on the diagonal. F
iii) An algorithm is a series of logical steps that solves one specific problem. τ
iv) The word true is a valid logical expression. τ
v) When moving a block of data in the form of specified rows and columns from array A to array B, the shape of the block in A must match the shape of the block in B. τ

2. Fill in the blanks: (1 × 5)

- i) Arithmetic operations can be performed collectively on the individual components of two arrays as long as both arrays ____ or one of them is ____.
ii) The normal precedence of operators can be overruled by the use of ____.
iii) A while loop can be repeated a number of times, depending on the ____ being processed.
iv) MATLAB provides a function called mesh to generate a mesh grid for 3D plotting.
v) The function used to find the determinant of a matrix in MATLAB is det(M).

3. Answer with proper justifications: (2 × 5)

- i) Find the indefinite integral of the function $f(x) = x^2 + 3x + 2$ using MATLAB's symbolic toolbox.
ii) Suppose X is a new variable, with the command, $X = [-10 : -1 : -15; -2 : 3]$. What will be the size of X?
iii) What is the difference between clc and clear commands?
iv) Determine the solution for the given system of equations using both matrix left division and the inverse matrix method:
$$\begin{aligned} 5x + 3y - z &= 10 \\ 3x + 2y + z &= 4 \\ 4x - y + 3z &= 12. \end{aligned}$$

v) Create a 3x3 matrix in MATLAB such that each element is the product of its row and column index.

4. Find the mistake in the following commands/codes, if any, and correct them: (1.5 × 6)

- i) $X[1,2] = 4$
ii) $P = \text{ones}(1;3)$
(1,3)

iii) `a = 5`
 `b = 3`
 `result = a + b;`
 `disp('The result is: ', result);`
 iv) `x = 1:5;`
 `y = x.^2;`
 `plot(x, y);`
 v) `a = 1;`
 `b = 2;`
 `c = a*b;`
 `print(c);`
 vi) `x=[1 3 5; 8 10 12]`
 `plot(x);`

5. Write code in MATLAB:

(3.5 × 6)

- i) Write a code to add the given numbers till the number entered is 0, using while loop.
- ii) Write a code using if statement to find the root of a quadratic equation $ax^2 + bx + c = 0$, where the inputs a , b , c are given by the user. Compute also the CPU time.
- iii) Write a code to find the 100th prime number.
- iv) Write a code to solve the following differential equation using MATLAB's Symbolic Math Toolbox:

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = 0,$$

with initial conditions $y(0) = 1$ and $y'(0) = 0$.

- v) Write a code to solve the initial value problem using 'ode45'

$$\frac{d^2y}{dt^2} + y = 0, \quad 0 < t < 5, \quad y(0) = 1, \quad y'(0) = 0.$$

- vi) Write a code to find the minimum value in an array of 100 numbers using for loop.

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